

The Fundamental Theorem of Calculus



Part 2: evaluating definite integrals

1 Evaluate each definite integral:

a $\int_1^3 (2x + 1) dx$

b $\int_0^\pi \sin x dx$

c $\int_1^4 \sqrt{x} dx$

2 Evaluate $\int_0^1 e^{2x} dx$, to three decimal places.

Part 1: differentiating an integral

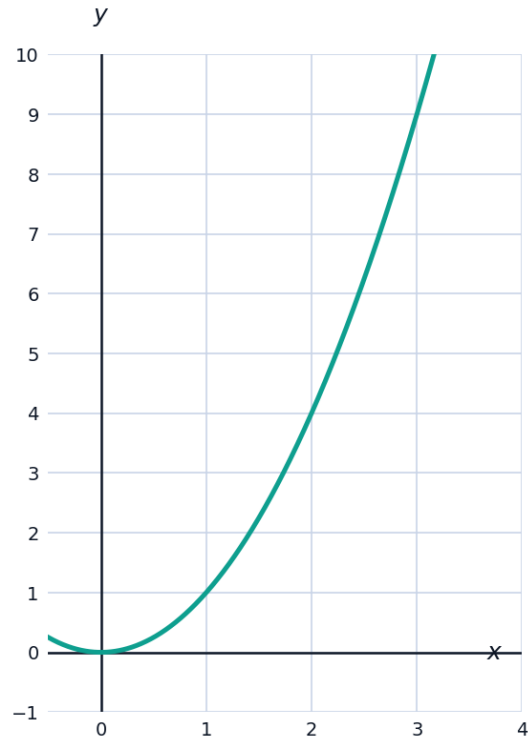
3 Use the Fundamental Theorem, Part 1, to find $\frac{d}{dx} \int_2^x (t^3 + 1) dt$.

4 Find $\frac{d}{dx} \int_1^{x^2} \sin(t) dt$ using the chain rule with FTC.

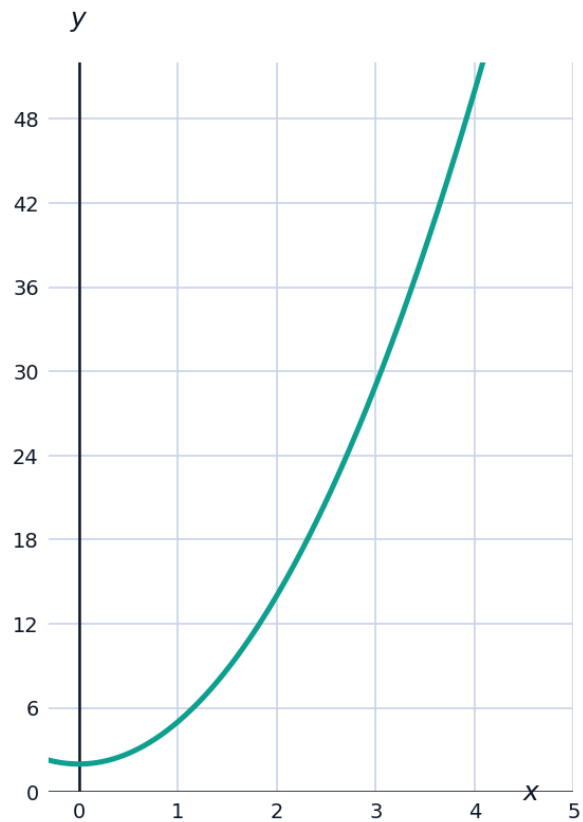
5 Let $g(x) = \int_0^x \sqrt{4 + t^2} dt$. Find $g'(3)$.

Average value and applications

- 6 Find the average value of $f(x) = x^2$ on $[0, 3]$.

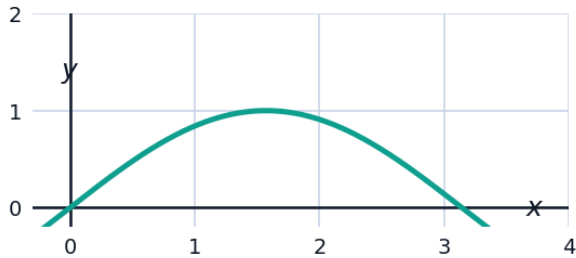


- 7 A car's velocity is $v(t) = 3t^2 + 2$ m/s for $0 \leq t \leq 4$. Find the total distance traveled, and the average velocity over this interval.



Further FTC practice

- 8 Evaluate $\int_0^2 (x^3 - 2x + 3) dx$.
- 9 Find $\frac{d}{dx} \int_x^5 \cos(t^2) dt$, using the property $\int_a^b = -\int_b^a$ before applying FTC Part 1.
- 10 Find the average value of $f(x) = \sin x$ on $[0, \pi]$, to three decimal places.

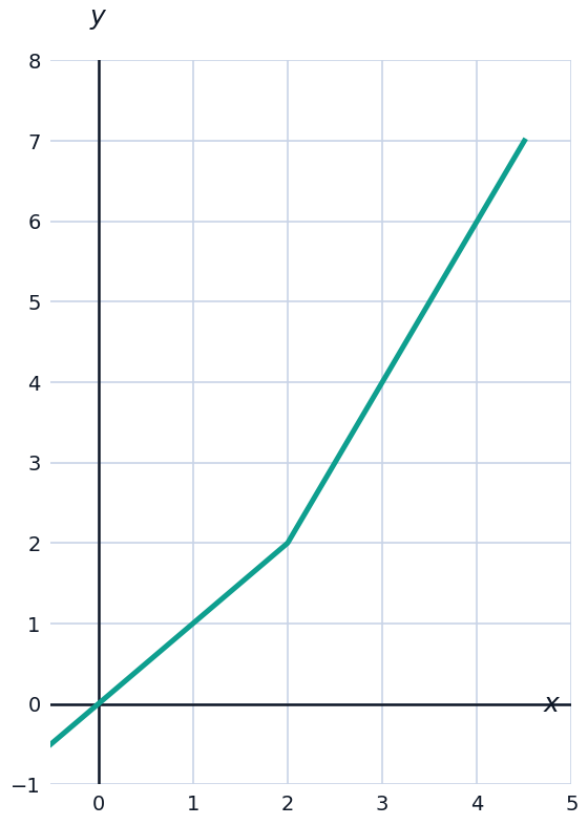


Mean Value Theorem for integrals

- 11 Find the value(s) of c guaranteed by the Mean Value Theorem for Integrals for $f(x) = x^2$ on $[0, 3]$ (where $f(c)$ equals the average value of f).
- 12 Evaluate $\int_2^5 f(x) dx$ given that the average value of f on $[2, 5]$ is 7.

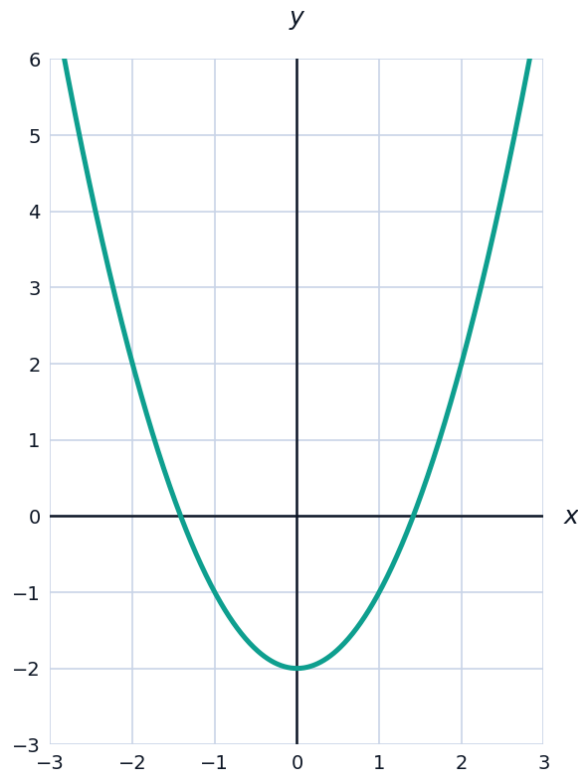
Integrals of piecewise functions

- 13 Evaluate $\int_0^4 f(x) dx$ where $f(x) = x$ for $x \leq 2$ and $f(x) = 2x - 2$ for $x > 2$.



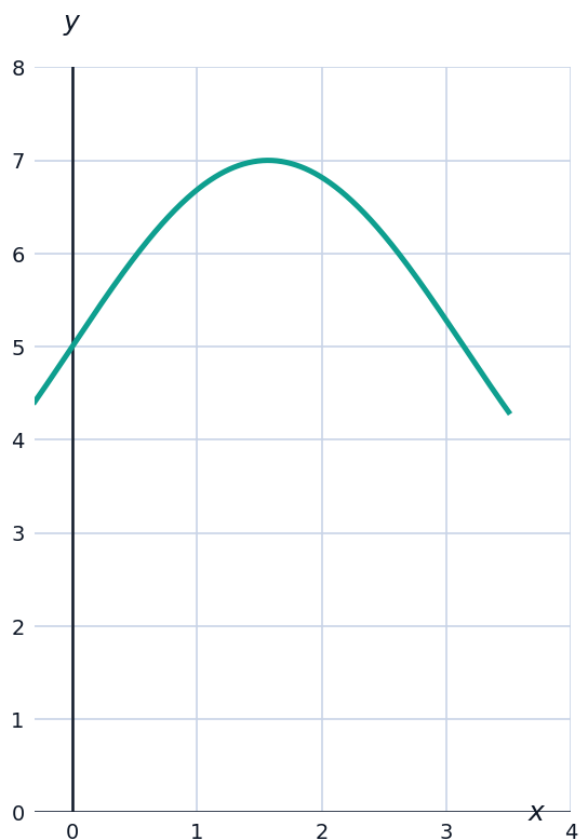
Visualizing the two parts of FTC together

- 14 For $g(x) = \int_0^x (t^2 - 2) dt$, find $g'(x)$ using FTC Part 1, then find where g has a local minimum by analyzing the sign of $g'(x)$.



A capstone FTC application

- 15 The rate of water flowing into a tank is $r(t) = 5 + 2\sin(t)$ liters/minute. Find the total volume that flows in during $[0, \pi]$ minutes, using FTC.

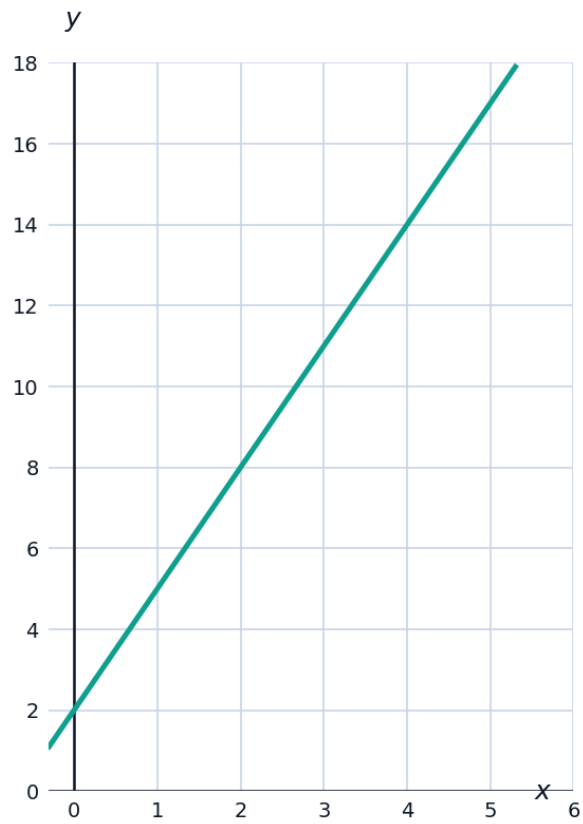


FTC Part 1 with more complex upper limits

- 16 Find $\frac{d}{dx} \int_0^{x^3} \sqrt{1+t^2} dt$ using the chain rule with FTC.
- 17 Find $\frac{d}{dx} \int_{x^2}^5 \cos(t) dt$, using $\int_{x^2}^5 = -\int_5^{x^2}$ before differentiating.

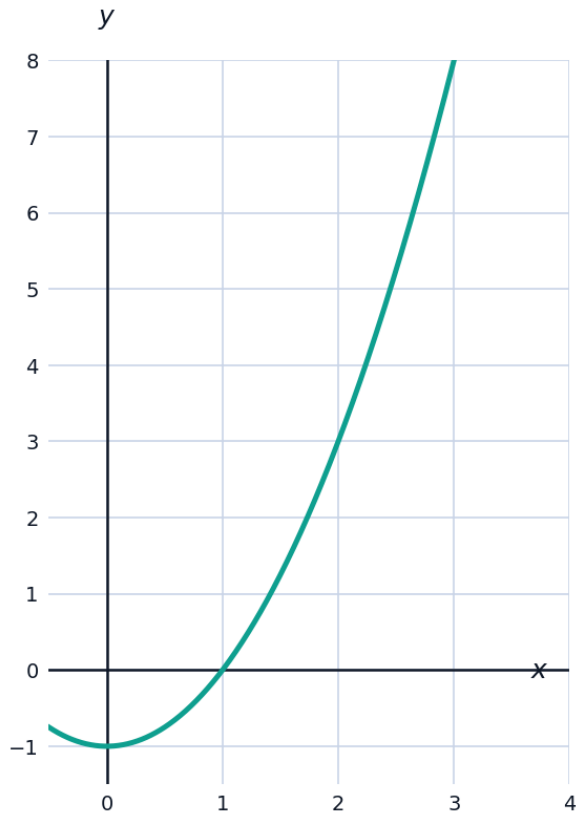
Net change and the FTC

- 18 A tank starts with 200 liters of water. Water flows out at rate $r(t) = 3t + 2$ liters/minute for the first 5 minutes. Use FTC to find the amount of water in the tank at $t = 5$.



Visualizing FTC Part 2 as area

- 19 The graph shows $f(x) = x^2 - 1$ on $[0, 3]$. Use FTC to find $\int_0^3 (x^2 - 1) dx$ and explain why the result is not simply the area of the shaded region (part of the curve dips below the axis).



Combining FTC Part 1 and Part 2 in one problem

- 20 Let $F(x) = \int_1^x (3t^2 - 2) dt$. Find $F(2)$ using Part 2, then verify $F'(2)$ using Part 1 and confirm it matches direct substitution into the integrand.